EfficientNet-GCN: A Hybrid Convolutional–Graph Framework for Explainable Multi-Class Osteoporosis Severity Classification from Knee Radiographs

Tasnim Sultana Sintheia, Musfique Rahman Muin, Pranta Das Gupta, and Md. Iftekharul Mobin

***Abstract*—Osteoporosis often remains undetected until a fragility fracture occurs, and limited access to dual-energy X-ray absorptiometry (DXA) leaves routine knee radiographs an underused screening opportunity in resource-constrained settings. Existing deep learning approaches to this task reason over the radiograph as a Euclidean grid, representing relationships between anatomically adjacent bone regions only implicitly through the convolutional receptive field, and typically expose a single coarse saliency map as their only account of a decision. This paper presents EfficientNet-GCN, a hybrid convolutional–graph framework for explainable three-class osteoporosis severity classification from knee radiographs. It comprises a *Patch-Graph Construction Module* that channel-reduces EfficientNetB0's spatial feature map and reformulates it as an eight-connected, 100-node patch graph; a *Residual Dual-Pooling Graph Reasoning Branch* that propagates relational context across three graph-convolutional layers and reads it out through concatenated mean–max pooling; and an *Auxiliary Convolutional Branch* fused with the graph branch by a weighted probability head. The 1,947-image corpus is partitioned before any augmentation is applied, and only the training partition is rebalanced to 3,000 images, so the validation and test partitions retain their natural class distribution throughout evaluation. On the 293-image held-out test set the model attains 82.94% accuracy (95% Wilson confidence interval 78.2–86.8%), a macro F1-score of 0.8161, and a macro one-vs-rest AUC of 0.9429, with 5.03M parameters. A three-tier explainability suite, Grad-CAM over the convolutional backbone, GNNExplainer over the patch graph, and LIME over super-pixels, localizes predictions to cortical and trabecular bone regions rather than background or soft tissue. The contribution of this work is a leakage-controlled evaluation protocol, an explicit relational representation over patch-level features, and the joint reporting of three complementary explanation views alongside a full parameter budget.**

***Index Terms*—Osteoporosis, bone mineral density, knee radiograph, EfficientNet, graph convolutional network, explainable artificial intelligence, Grad-CAM, opportunistic screening, medical image classification.**

I. Introduction

OSTEOPOROSIS is a progressive systemic skeletal disorder characterized by reduced bone mineral density (BMD) and deterioration of bone microarchitecture, resulting in fragility fractures [1]. It is often described as a silent disease because it produces no symptoms until a fracture occurs, at which point it causes ongoing pain, loss of independence, and increased mortality, particularly among the elderly [2]. As the global population ages, the incidence of osteoporosis is expected to rise, placing a significant socio-economic burden on healthcare systems worldwide. Dual-energy X-ray absorptiometry (DXA) is the diagnostic gold standard, but the cost of the equipment, its concentration in tertiary centres, and the referral pathway it requires together prevent its use as a population-level screening tool [3]. Timely osteoporosis screening therefore remains scarce, especially in resource-poor and rural healthcare settings [4].

Conventional knee radiography, by contrast, is inexpensive, widely available, and routinely acquired during musculoskeletal assessment for osteoarthritis and trauma [5]. Radiographs already acquired for these indications therefore present an opportunity for opportunistic osteoporosis screening without additional acquisitions or additional radiation exposure [3]. Manual assessment of trabecular pattern and cortical appearance on knee radiographs is, however, difficult and subject to high inter-observer variability, which limits its usefulness for early diagnosis [5]. Recent advances in deep learning, and Convolutional Neural Networks (CNNs) in particular, have shown considerable promise for automatically learning the complex imaging patterns and sub-visual bone texture features associated with bone degeneration [5], [6].

Although CNN architectures such as ResNet and VGG have demonstrated strong performance on this task [5], several limitations remain. Current approaches extract local spatial features within fixed receptive fields and do not explicitly model relationships between anatomically related regions of the knee across scales [7]. Most existing methods are additionally black boxes, offering little interpretability and thereby reducing clinician confidence in the resulting diagnostic decisions [8]. These drawbacks point to a need to combine discriminative feature extraction, structural reasoning, and clinically meaningful explainability within a single framework [7], [8].

To address these problems, this study proposes EfficientNet-GCN, a hybrid deep learning framework for the explainable classification of osteoporosis severity from knee radiographs. The framework integrates the feature extraction capacity of

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EfficientNet with the relational reasoning capacity of Graph Convolutional Networks (GCNs), representing relationships between knee-joint structures explicitly rather than treating each spatial location in isolation. Using a multi-class knee radiograph corpus of 1,947 images [9], the model classifies radiographs into three clinically relevant bone-health categories: Normal, Osteopenia, and Osteoporosis. Embedded explainability techniques provide transparency into which image regions drive each prediction. The principal contributions of this work are as follows.

- A **Patch-Graph Construction Module** that channel-reduces and reformulates EfficientNetB0's $1280 \times 10 \times 10$ spatial feature map into an eight-connected, 100-node patch graph, providing an explicit relational representation over anatomically adjacent knee regions (Section III-G).
- A **Residual Dual-Pooling Graph Reasoning Branch** comprising three stacked graph-convolutional layers with residual connections around the second and third layers, read out by a concatenated mean–max pooling operator that retains both a joint-wide density summary and the response of the single most abnormal patch (Section III-H).
- An **Auxiliary Convolutional Branch and Weighted Ensemble Head** that fuses graph-based and Euclidean predictions through a 0.6/0.4 probability combination, providing a graph-free path from features to logits alongside the relational one (Section III-I).
- A **bone-aware preprocessing and class-balancing pipeline** that applies Contrast Limited Adaptive Histogram Equalization (CLAHE) at native resolution and rebalances the training partition by targeted augmentation, correcting the approximately two-to-one class imbalance without distorting fine trabecular detail (Sections III-B and III-D).
- A **leakage-controlled evaluation protocol** in which the corpus is partitioned before any augmentation is applied and only the training partition is rebalanced, so that no augmented variant of a test radiograph can appear in training (Section III-C).
- A **three-tier explainability suite** combining Grad-CAM, GNNExplainer, and Local Interpretable Model-agnostic Explanations (LIME) to examine, at three levels of granularity, whether predictions are grounded in cortical and trabecular bone regions (Section III-M).

The remainder of this paper is organized as follows. Section II reviews related work and closes by synthesizing the research gaps this study addresses. Section III describes the methodology. Section IV presents results. Section V discusses their interpretation and the framework's principal methodological vulnerability. Sections VI and VII state limitations and future work, and Section VIII concludes.

II. Related Work

This section reviews recent literature across four themes: CNN and transfer-learning approaches specific to knee osteoporosis, opportunistic screening from other skeletal sites, graph-based relational learning in medical imaging, and explainability in bone-health assessment. It closes by synthesizing the gaps that motivate the present study.

A. CNN and Transfer-Learning Approaches to Knee Osteoporosis Classification

Deep learning has become increasingly prominent in automated osteoporosis classification, with recent systematic reviews and meta-analyses reporting good diagnostic performance across a range of imaging datasets [10]–[12]. Most existing work adopts transfer learning with CNN backbones such as VGG [13], ResNet [14], Inception [15], and Xception [16] for binary and multi-class osteoporosis classification [5], [17]–[19]. Wani and Arora [5] compared four pretrained backbones on a small knee-radiograph corpus and found AlexNet the strongest of the group, illustrating how sensitive Euclidean CNN backbones are to the particular architecture–dataset pairing, while Sarhan et al. [19] later reported 97.50% accuracy on the same 1,947-image knee corpus used in the present study with VGG19, ResNet50, and a custom CNN under heavy data augmentation. More recent single-backbone efforts continue this trend: Muzaffar et al. [20] introduced OsteoNet, an attention-augmented VGG network for bone radiographs; Salunke et al. [21] paired VGG-16 features with a logistic-regression classifier on combined radiograph and BMD data; and Kaur et al. [22] concatenated MobileNetV2 and NasNet representations after rolling-guidance-filter enhancement of knee radiographs. The compound-scaled EfficientNet family [23] adopted in the present work was designed specifically to balance network depth, width, and input resolution under a fixed computational budget, which motivates its use here over the older, more parameter-heavy backbones surveyed above. Other studies apply segmentation methods such as U-Net [24] to isolate regions of interest prior to classification [17], [18]; this can localize the analysis but, like manual cropping, risks discarding the anatomical context surrounding the region of interest, while conventional data augmentation alone can aid generalization without necessarily improving the trabecular contrast the network must learn from [19], [25]. Across this body of work, a consistent property is that every model reasons over the image as a Euclidean grid: neighbouring bone regions influence the prediction only implicitly, through the receptive field of the convolution, never through an explicit relational representation.

B. Opportunistic Osteoporosis Screening Beyond the Knee

A broader body of work pursues opportunistic osteoporosis screening from imaging acquired for unrelated clinical indications. Hsieh et al. [26] trained a deep network on pelvis and lumbar-spine radiographs to jointly predict BMD and fracture risk, reporting 91.7% accuracy for hip osteoporosis and 86.2% for spine osteoporosis. Jang et al. [27] and Tseng et al. [28] instead repurposed chest radiographs, a far more ubiquitous examination, to flag low BMD without a dedicated skeletal view, while Zhang et al. [29] and Yamamoto et al. [30] demonstrated multicentre CNN screening from lumbar-spine and hip radiographs respectively, each combining imaging features with clinical covariates. Zhou et al. [31] recently extended this line to biplanar X-ray radiography, fusing anteroposterior and lateral views for bone-density prediction, and Kuo et al. [32] paired a Vision Transformer with a CNN to make DXA-referral recommendations directly from low-dose chest CT, explicitly optimizing for decision transparency. At the higher-cost end of the acquisition spectrum, opportunistic BMD estimation has also been explored from routine CT [33], [34] and conventional MRI [35], both of which trade wider availability for substantially higher acquisition cost than plain radiography. Collectively, this literature confirms that the underlying premise of the present study, that images acquired for other clinical purposes carry a usable BMD signal, is well supported, while also highlighting that knee radiographs remain comparatively under-explored relative to the hip, spine, and chest.

C. Graph-Based Relational Learning in Medical Imaging

While CNNs are efficient, they mainly encode local spatial representations through convolution and largely overlook long-range relational dependencies between learned image features [10], [19]. Graph Convolutional Networks [36] and their attention-based variants [37] were developed precisely to model such non-Euclidean, relational dependencies and have since been surveyed extensively across data-mining and vision applications [38]; Mienye and Viriri [7] review their emerging role in medical imaging specifically, where graph-structured reasoning over anatomical or histological regions has shown promise for capturing relationships that fixed-receptive-field convolutions miss. Most directly relevant to the patch-graph design adopted here, Ye et al. [39] propose MedGNN, which segments a medical image into patches, treats each patch as a node connected by k-nearest-neighbour edges, and applies a multi-scale graph-convolutional network for recognition, reporting consistent gains over grid- and sequence-based CNN and Transformer baselines across several imaging modalities. The present work adopts a similar patch-to-graph construction but fixes an eight-connectivity spatial adjacency rather than learning a k-nearest-neighbour graph, which adds no adjacency parameters at all; the trade-offs of this choice are examined critically in Section V. Despite this progress, only a small number of studies apply graph-based relational learning specifically to osteoporosis assessment, leaving the anatomical relationships between adjacent knee regions largely unexploited by the CNN-only literature surveyed above.

D. Explainability in Automated Bone-Health Assessment

The study of explainability in this domain remains comparatively underdeveloped. Coarse heatmaps generated by Grad-CAM [40] are commonly used to illustrate discriminative regions [6], [17] but do not offer detailed, instance-specific information about which image regions drove an individual prediction, which limits clinical interpretability. Complementary post-hoc techniques such as LIME [41], SHAP [42], and, for graph-structured predictions specifically, GNNExplainer [43] address different facets of this transparency gap, but their joint application to osteoporosis screening has not been systematically explored.

E. Synthesis: Research Gaps Addressed in This Work

Table I consolidates the studies discussed above, reporting for each the imaging modality and cohort size, the number of classes, the headline result, the parameter count where available, and whether any explainability method was reported. Four gaps emerge, each mapping onto a contribution listed in Section I.

- **Gap 1 – Euclidean-only feature learning.** Every CNN-based method in Table I reasons over the image as a fixed grid, representing relationships between anatomically adjacent bone regions only implicitly [7], [10], [19]. *Addressed by* the Patch-Graph Construction Module and the Residual Dual-Pooling Graph Reasoning Branch (Sections III-G–III-H).
- **Gap 2 – Limited interpretability.** As the narrative review above indicates, reviewed models are typically either black boxes or rely on a single coarse saliency method [8], [40]; the final column of Table I records what each study reports. *Addressed by* the three-tier explainability suite (Section III-M).
- **Gap 3 – Dataset imbalance.** Knee-radiograph corpora, including the one used here, are naturally imbalanced across severity classes, and several reviewed studies train directly on the raw distribution [5], [25]. *Addressed by* the bone-aware preprocessing and class-balancing pipeline (Section III-D).
- **Gap 4 – Single-branch fragility.** Reviewed methods typically commit to a single representation, either purely convolutional or purely graph-based, with no mechanism to hedge against the failure modes specific to either view. *Addressed by* the Auxiliary Convolutional Branch and Weighted Ensemble Head (Section III-I).

Two further gaps are visible in Table I but are *not* resolved by the present work, and are carried forward explicitly to Section VI rather than claimed as contributions. **Gap 5 – No external validation:** nearly all reviewed studies, including this one, evaluate on a single institutional dataset without multi-site validation. **Gap 6 – Missing efficiency reporting:** few reviewed studies report model size alongside accuracy, which makes the practical deployability of competing approaches difficult to compare; this is visible in the *N/R* entries of Table I. The present model's parameter count is reported in full in Section IV-G, but this does not resolve the gap in the wider literature.

III. Methodology

The study followed a five-stage pipeline: dataset curation, radiographic preprocessing, class-balancing augmentation, hybrid convolutional–graph feature learning, and post-hoc explainability. The overall workflow is illustrated in Fig. 1.

A. Dataset Description

The dataset consists of 1,947 knee radiographs distributed across three diagnostic categories: Normal (780 images), Osteopenia (374 images), and Osteoporosis (793 images) [9]. The imbalance is notable, with Osteopenia accounting for

TABLE I

CONSOLIDATED COMPARISON OF DEEP-LEARNING APPROACHES TO OSTEOPOROSIS AND BONE-DENSITY ASSESSMENT. *N/R* DENOTES A QUANTITY EXPLICITLY NOT REPORTED BY THE SOURCE STUDY; **[TBD]** DENOTES A VALUE STILL TO BE EXTRACTED FROM IT.

Study (Year)	**Modality (n)**	**Cls.**	**Reported result**	**Params (M)**	**XAI reported**	**Key limitation**
Zhang et al. [29] (2020)	Lumbar X-ray (2,510)	2	AUC 0.77–0.81	N/R	**[TBD]**	Modest discrimination
Yamamoto et al. [30] (2020)	Hip X-ray	**[TBD]**	**[TBD]**	N/R	**[TBD]**	Requires clinical covariates
Hsieh et al. [26] (2021)	Pelvis/lumbar X-ray (>5,000)	2	91.7% / 86.2% acc.	N/R	**[TBD]**	Binary formulation
Sukegawa et al. [6] (2022)	Panoramic dental (778)	2	84.50% acc.	N/R	**[TBD]**	Manual ROI cropping
Jang et al. [27] (2022)	Chest X-ray	**[TBD]**	**[TBD]**	N/R	**[TBD]**	Binary formulation
Wani & Arora [5] (2023)	Knee X-ray (381)	3	91.10% acc.	N/R	**[TBD]**	Very small cohort
Feng et al. [18] (2024)	Hip X-ray (139)	3	74.00% acc.	N/R	**[TBD]**	Extremely small dataset
Naguib et al. [25] (2024)	Knee X-ray (611)	3	83.74% acc.	N/R	**[TBD]**	High model complexity
Sarhan et al. [19] (2024)	Knee X-ray (1,947)	3	97.50% acc.	N/R	**[TBD]**	Augmentation applied before partitioning
Chen et al. [17] (2024)	Hip X-ray (3,460)	Multi + T-score	97.20% acc.	N/R	**[TBD]**	Implant images excluded
Galbusera et al. [35] (2024)	Lumbar MRI/X-ray (429)	2	AUC 0.80–0.82	N/R	**[TBD]**	Higher-cost modality
Park et al. [33] (2024)	CT (422)	2	AUC 0.77–0.79	N/R	**[TBD]**	Higher-cost modality
Muzaffar et al. [20] (2025)	Bone radiographs	**[TBD]**	N/R	N/R	**[TBD]**	Single backbone, no relational modelling
Kuo et al. [32] (2025)	Low-dose chest CT	**[TBD]**	**[TBD]**	N/R	**[TBD]**	Higher-cost modality
Kaur et al. [22] (2026)	Knee X-ray	**[TBD]**	N/R	N/R	**[TBD]**	No graph-based reasoning
Proposed (this work)	**Knee X-ray (1,947)**	**3**	**82.94% acc., AUC 0.9429**	**5.03**	**Grad-CAM, GNNExplainer, LIME**	**Single-centre; ablation incomplete**

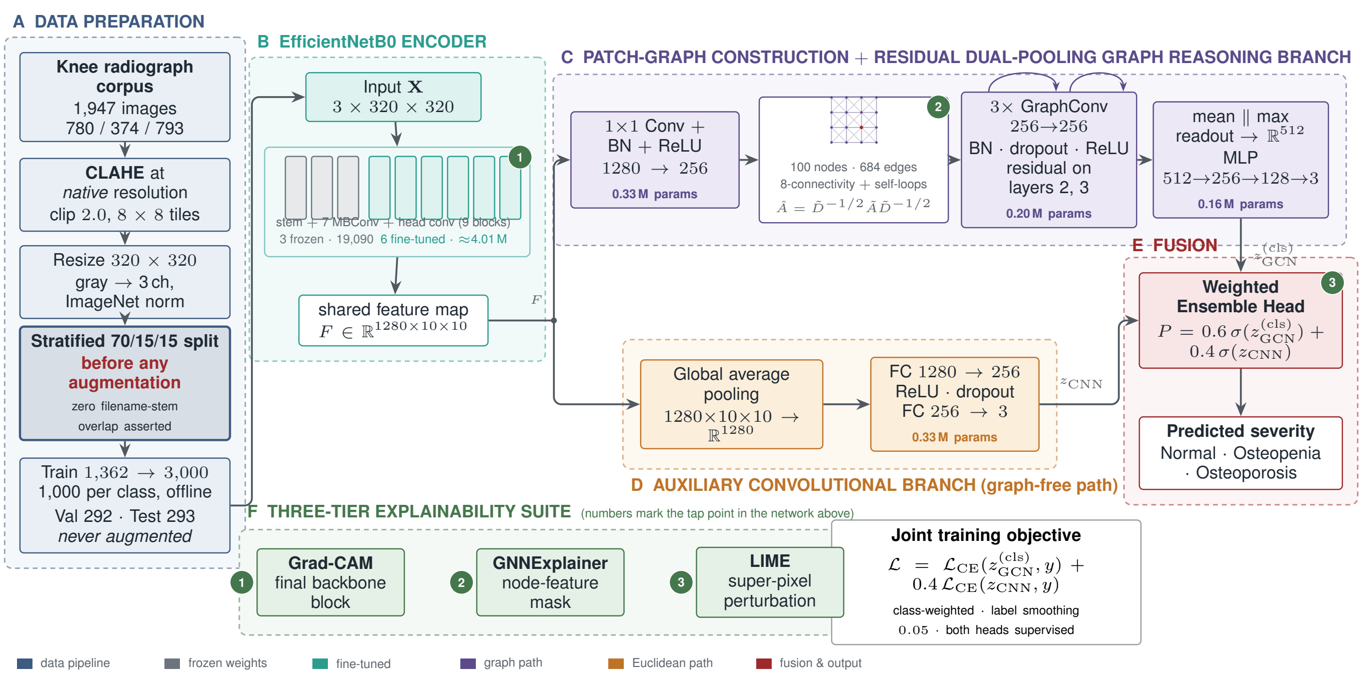

Fig. 1. End-to-end EfficientNet-GCN pipeline. (A) The corpus is contrast-enhanced at native resolution and partitioned *before* any augmentation, so no augmented variant of a validation or test radiograph can reach training. (B) An EfficientNetB0 trunk with its stem and first two MBConv stages frozen produces the shared feature map F. (C) F is channel-reduced and reformulated as a 100-node, eight-connected patch graph that three residual graph-convolutional layers propagate over, read out by concatenated mean–max pooling. (D) In parallel, a graph-free branch pools F directly. (E) The two class-probability vectors are fused $0.6/0.4$. (F) Three explanation methods tap the network at the numbered points. Parameter counts are given per component; the model totals 5.03 M.

less than a fifth of the corpus, a disparity that shaped the augmentation strategy of Section III-D. Images were organized by class into separate directories and read with a standard folder-labelled loader, so class membership was inferred from directory structure rather than an external annotation file.

The corpus is distributed as a public Kaggle dataset. Several properties relevant to methodological interpretation are not documented by the source and could not be recovered: the acquiring institution or institutions, patient demographics, acquisition parameters, the qualifications of the annotators,

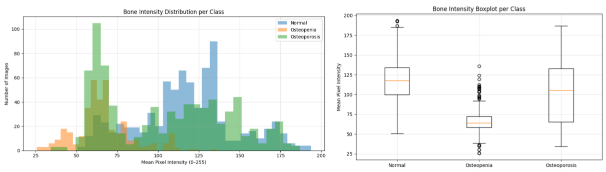

Fig. 2. Bone intensity distribution per class.

and whether the severity labels were established against DXA-derived T-scores or by radiographic reading alone. Critically, the distribution provides no patient-level identifiers, so it cannot be determined whether any patient contributed more than one radiograph, for example a bilateral pair or a follow-up study. The consequences of these unknowns for the evaluation protocol are addressed directly in Sections III-C and VI.

B. Data Preprocessing

Every image was first converted to grayscale, and CLAHE [44] (clip limit 2.0, tile grid 8×8) was applied *at the image's native resolution*, before any resizing. This ordering is deliberate: enhancing contrast first and resizing second preserves the fine trabecular striations that carry much of the diagnostic signal in bone-density assessment, whereas resizing first and enhancing contrast afterwards would let CLAHE amplify resampling artefacts rather than genuine bone structure. No denoising filter is applied, since a low-pass step before resizing would smooth away the same trabecular texture. The contrast-enhanced image is then resized to 320×320 pixels using area interpolation when downscaling or cubic interpolation when upscaling, and the single grayscale channel is replicated across three channels to match the input expectations of an ImageNet-pretrained backbone.

An input resolution of 320×320 was chosen rather than EfficientNetB0's native 224×224 for two reasons: it preserves more of the high-frequency trabecular detail that CLAHE is applied to enhance, and it yields a 10×10 terminal feature map, giving a 100-node patch graph whose spatial granularity is fine enough to distinguish sub-regions of the joint while remaining small enough to keep the graph branch inexpensive. The effect of this choice has not been ablated (Section VI).

The choice of enhancement parameters was informed by a preliminary bone-intensity analysis across the three classes, computing the mean and standard deviation of grayscale pixel intensity per class (Fig. 2). Class-wise means were **[TBD]** (Normal), **[TBD]** (Osteopenia), and **[TBD]** (Osteoporosis), with standard deviations **[TBD]**, **[TBD]**, and **[TBD]** respectively; a Kruskal–Wallis test across the three groups gave **[TBD]**. This analysis surfaced a measurable, if modest, intensity gradient across the classes, which supported the use of a local, tile-based contrast method rather than a single global histogram stretch. At load time, images are normalized using standard ImageNet statistics (mean $0.485, 0.456, 0.406$; standard deviation $0.229, 0.224, 0.225$).

C. Dataset Split

Unlike a design that augments first and partitions second, the split here is performed *before* any augmentation, directly

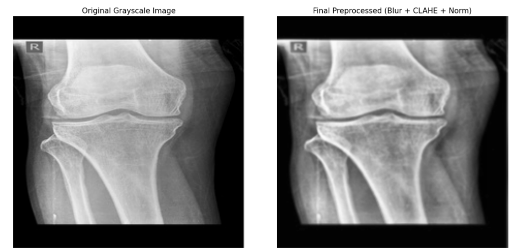

Fig. 3. Original versus preprocessed radiograph (CLAHE at native resolution, then resized).

TABLE II
DATASET DISTRIBUTION: ORIGINAL 70/15/15 SPLIT, THEN TRAINING-ONLY AUGMENTATION

Class	Train (orig.)	Val	Test	Train (aug.)
Normal	546	117	117	1000
Osteopenia	261	56	57	1000
Osteoporosis	555	119	119	1000
Total	**1362**	**292**	**293**	**3000**

on the preprocessed originals: each class is divided 70/15/15 into training, validation, and test partitions (Table II), yielding 1,362 training, 292 validation, and 293 test images. This ordering matters methodologically. Augmenting before splitting can place near-duplicate variants of the same source radiograph in both the training and test partitions, so that reported accuracy partly reflects memorization of a specific radiograph rather than generalization to an unseen one. Splitting first guarantees that every source radiograph appears in exactly one partition; this was verified programmatically by asserting zero filename-stem overlap between the train, validation, and test directories after the split, and again after augmentation.

This check establishes that no *image* appears in more than one partition. It does not establish that no *patient* does, because the source distribution carries no patient identifiers (Section III-A). If the corpus contains bilateral pairs or repeat studies from the same individual, an image-level partition can still leak patient identity across partitions, and the reported test metrics would be correspondingly optimistic. This residual risk cannot be excluded with the metadata available and is restated as a limitation in Section VI.

Only the training partition is subsequently rebalanced by augmentation (Section III-D); the validation and test partitions are left at their natural, imbalanced class distribution and are evaluated using only the deterministic preprocessing of Section III-B, with every metric reported as a macro average (Section III-L) precisely because the test set is not class-balanced.

D. Class-Balancing Augmentation

Given the roughly two-to-one disparity between the smallest and largest classes in the training partition, training directly on the raw distribution risked biasing the classifier toward the majority classes. Rather than reweighting the loss alone, the training partition itself was rebalanced through targeted, offline augmentation, applied only after the split of Section III-C and only to the training images. Every original training image is retained unchanged, and each class is then topped up with augmented variants, cycling back through the originals as needed, until it reaches 1,000 images. An Albumentations [45] pipeline supplies the augmented variants, drawing from horizontal flip ($p = 0.5$), a combined affine transform with scaling in $[0.90, 1.10]\times$, translation up to 6% per axis, and rotation within $\pm 12^\circ$ ($p = 0.8$), brightness/contrast jitter ($p = 0.5$), and additive Gaussian noise ($p = 0.3$). Each transform is applied independently at its own probability rather than exactly one per image, so an augmented sample may combine several perturbations. This step brings the training partition to 1,000 images per class, 3,000 in total (Table II); the validation and test partitions are never augmented.

A second, online augmentation stage is applied on top of this balanced training set at every training step, implemented with torchvision transforms: random resized cropping (scale $[0.85, 1.0]$), random horizontal flip, random rotation up to 10°, brightness/contrast jitter, and random erasing ($p = 0.25$). Random erasing is retained at a low probability as a mild occlusion-robustness measure; because it can in principle occlude diagnostically relevant bone regions, its net effect on this task has not been isolated and is listed among the unablated design choices in Section VI. Hue and saturation jitter are deliberately excluded, since the radiographs are grayscale replicated across three channels and colour perturbations would desynchronize the channels into hues that cannot occur in a real radiograph. Validation and test images instead pass through a fixed, deterministic pipeline of resizing to 320×320 followed by ImageNet normalization only, with no stochastic transform of any kind.

E. Framework Overview and Forward Pass

EfficientNet-GCN is organized around three named components, described in turn in Sections III-F–III-I: a **Patch-Graph Construction Module** that channel-reduces and turns the backbone's spatial feature map into a relational patch graph, a **Residual Dual-Pooling Graph Reasoning Branch** that propagates information across that graph and reads it out, and an **Auxiliary Convolutional Branch** whose prediction is fused with the graph branch's prediction by a **Weighted Ensemble Head**. Fig. 1 summarizes the resulting dataflow.

Let $\mathbf{X} \in \mathbb{R}^{3\times 320\times 320}$ denote a preprocessed input image, $E_\theta(\cdot)$ the EfficientNetB0 encoder of Section III-F, $\mathrm{Proj}(\cdot)$ the channel-reduction step and $\mathrm{PatchGraph}(\cdot)$ the graph-construction operator of Section III-G, $\mathrm{GCN}_3(\cdot)$ and $\mathrm{DualPool}(\cdot)$ the three-layer graph branch and its readout from Section III-H, and $\mathrm{MLP}_{\text{graph}}(\cdot)$ and $\mathrm{MLP}_{\text{aux}}(\cdot)$ the two branches' classification heads. The forward pass can then be written compactly as

$$
\begin{aligned}
F &= E_\theta(\mathbf{X}),\\
F' &= \mathrm{Proj}(F), \qquad \mathcal{G} = \mathrm{PatchGraph}(F'),\\
z_{\text{GCN}} &= \mathrm{DualPool}\big(\mathrm{GCN}_3(\mathcal{G})\big),\\
z_{\text{GCN}}^{(\text{cls})} &= \mathrm{MLP}_{\text{graph}}(z_{\text{GCN}}),\\
z_{\text{CNN}} &= \mathrm{MLP}_{\text{aux}}\big(\mathrm{GAP}(F)\big),\\
P_{\text{final}} &= 0.6\,\mathrm{softmax}\big(z_{\text{GCN}}^{(\text{cls})}\big) + 0.4\,\mathrm{softmax}\big(z_{\text{CNN}}\big),
\end{aligned}
\tag{1}
$$

where $F \in \mathbb{R}^{1280\times 10\times 10}$ is the spatial feature map shared by both branches, $F' \in \mathbb{R}^{256\times 10\times 10}$ is its channel-reduced counterpart, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is the 100-node patch graph of Section III-G, and $P_{\text{final}} \in \mathbb{R}^3$ is the final class-probability vector reported in Section IV. All tensor shapes follow the channels-first convention. Throughout, $\phi(\cdot)$ denotes the ReLU nonlinearity and $\mathrm{softmax}(\cdot)$ the softmax function; the two are kept notationally distinct because both appear as activation functions in different parts of the model.

The two branches deliberately consume different tensors: the graph branch reasons over the channel-reduced F', while the auxiliary branch pools the full, un-reduced F directly. This shared-backbone, dual-branch design lets the graph branch reason relationally over patch neighbourhoods while the auxiliary branch retains a direct, graph-free path from features to logits; the two views are combined at the output by Eq. (6).

F. EfficientNetB0 Radiographic Encoder

The shared encoder $E_\theta(\cdot)$ is EfficientNetB0 [23], initialized from ImageNet weights. Its convolutional trunk consists of a stem followed by seven MBConv stages and a final 1×1 head convolution, nine top-level blocks in total. Rather than freezing the entire network or fine-tuning it wholesale, only the last six of these nine blocks, the final five MBConv stages together with the head convolution, were made trainable, leaving the stem and the first two MBConv stages frozen at their ImageNet-pretrained weights; this compromise lets the network adapt its higher-level filters to radiographic texture while leaving the most generic low-level filters undisturbed. For an input of 320×320, the backbone produces the spatial feature map $F \in \mathbb{R}^{1280\times 10\times 10}$ that feeds both downstream branches.

G. Patch-Graph Construction Module

Before graph construction, a 1×1 convolution channel-reduces the shared feature map from 1,280 to 256 channels:

$$
F' = \phi\big(\mathrm{BN}(\mathrm{Conv}_{1\times 1}(F))\big) \in \mathbb{R}^{256\times 10\times 10}. \tag{2}
$$

This projection exists to control the graph branch's parameter count: a graph-convolutional layer consuming the raw 1,280-channel features would require a 1280×1280 weight matrix, roughly 1.64M parameters in a single layer, whereas the 1×1 convolution performs the reduction with 0.33M parameters and leaves each subsequent graph layer at 256×256, or 65.5K parameters. Rather than pooling F' directly, as a conventional CNN classifier would, the Patch-Graph Construction Module treats each of the one hundred spatial positions of F' as a node $x_i \in \mathbb{R}^{256}$ of a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $|\mathcal{V}| = 100$ (Fig. 4, panel A). Nodes are connected to their spatial neighbours under an eight-connectivity scheme, so each patch is linked to its orthogonal and diagonal neighbours alike (Fig. 4, panel B), producing 684 directed edges over the 10×10 grid. The resulting adjacency A is made self-connected and symmetrically normalized before being consumed by the graph branch (Fig. 4, panel C):

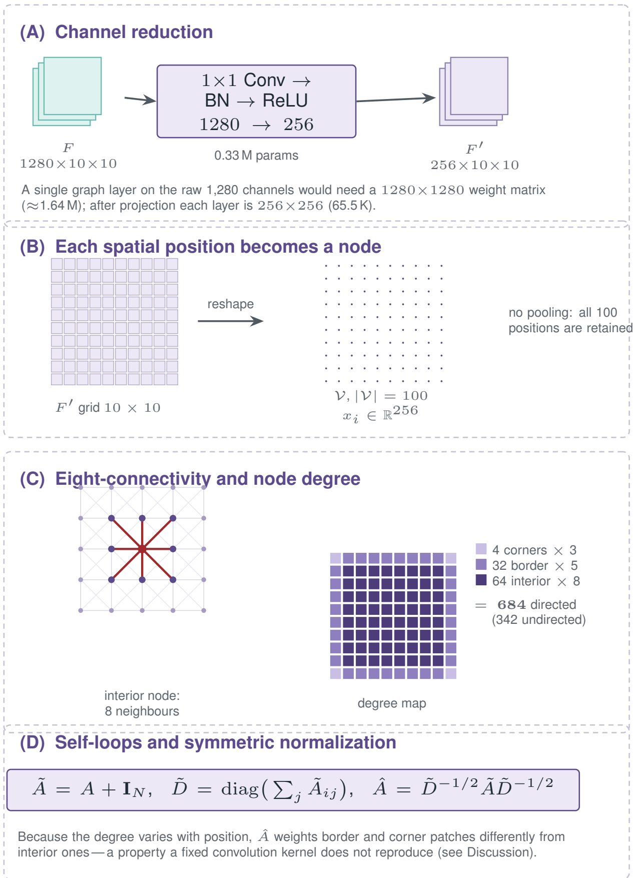

Fig. 4. Patch-Graph Construction Module. (A) A 1×1 convolution reduces the shared feature map from 1,280 to 256 channels (Eq. (2)), which keeps each subsequent graph layer at 256×256 rather than 1280×1280. (B) Each of the 100 spatial positions becomes one node $x_i \in \mathbb{R}^{256}$; no pooling is applied. (C) Eight-connectivity links every patch to its orthogonal and diagonal neighbours. The degree map shows why the edge count is 684: 4 corners of degree 3, 32 border patches of degree 5, and 64 interior patches of degree 8. (D) Self-loops are added and the adjacency is symmetrically normalized (Eq. (3)); because degree varies with position, $\hat{A}$ weights border and interior patches differently.

$$\tilde{A} = A + \mathbf{I}_N, \qquad \tilde{D} = \mathrm{diag}\Big(\textstyle\sum_j \tilde{A}_{ij}\Big), \qquad \hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}, \tag{3}$$

where $\mathbf{I}_N$ is the $N \times N$ identity matrix with $N = 100$ and $\tilde{D}$ is the degree matrix of the self-looped adjacency $\tilde{A}$. The node set $\{x_i\}_{i=1}^{100}$ together with the normalized adjacency $\hat{A}$ constitutes the graph $\mathcal{G}$ consumed by the Residual Dual-Pooling Graph Reasoning Branch of Section III-H.

A fixed adjacency was chosen over an attention-weighted alternative such as GAT [37] or a learned k-nearest-neighbour construction [39] because the neighbourhood structure of a regular grid is known a priori and requires no parameters to represent. This design choice has a significant analytical consequence, since the graph is then identical for every input image; that consequence is examined in Section V.

H. Residual Dual-Pooling Graph Reasoning Branch

The patch graph $\mathcal{G}$ is processed by three stacked graph-convolutional layers, all $256 \to 256$ since the channel reduction was already performed by the Patch-Graph Construction Module (Eq. (2)), each following the first-order localized spectral propagation rule of Kipf and Welling [36] (Fig. 5):

$$H^{(l+1)} = \phi\left(\hat{A}\, H^{(l)} W^{(l)}\right), \tag{4}$$

where $\hat{A}$ is the normalized self-looped adjacency of Eq. (3), $H^{(l)}$ denotes the node features entering layer l with $H^{(0)}$ the node set $\{x_i\}$, $W^{(l)}$ is the layer's learnable weight matrix, and ϕ is the ReLU nonlinearity. Each layer is followed by batch normalization, dropout, and ϕ, with residual connections added around the second and third layers to ease optimization as depth increases. Rather than pooling the graph with a single readout, the branch concatenates a global mean-pooling operation, capturing the overall density signal across the joint, with a global max-pooling operation, capturing the single most abnormal patch:

$$z_{\text{GCN}} = \left[\mathrm{MeanPool}\big(H^{(3)}\big)\,;\, \mathrm{MaxPool}\big(H^{(3)}\big)\right] \in \mathbb{R}^{512}. \tag{5}$$

This 512-dimensional graph embedding then passes through a three-layer fully connected head ($512 \to 256 \to 128 \to 3$) to produce the graph branch's class logits $z^{(\text{cls})}_{\text{GCN}}$.

I. Auxiliary Convolutional Branch and Weighted Ensemble Head

A second, considerably simpler branch operates in parallel on the same shared feature map F, bypassing the graph entirely (Fig. 6): global average pooling followed by two fully connected layers ($1280 \to 256 \to 3$) with a ReLU activation and dropout in between, producing auxiliary logits z_{CNN}. This branch is trained jointly with the graph branch, and the two predictions are combined at inference by the Weighted Ensemble Head:

$$P_{\text{final}} = w \cdot \mathrm{softmax}(z^{(\text{cls})}_{\text{GCN}}) + (1-w)\cdot \mathrm{softmax}(z_{\text{CNN}}), \quad w = 0.6. \tag{6}$$

The larger weight is assigned to the graph branch on the grounds that it has access to inter-patch relationships the auxiliary branch does not. The specific value $w = 0.6$ was fixed a priori rather than tuned, and was set to mirror the auxiliary loss weight of Eq. (7) for simplicity; no sensitivity analysis over w has yet been performed, and the sweep required to justify the value empirically is reported as **[TBD]** pending completion (Section VI).

Because both heads share a single backbone and are optimized jointly through the coupled objective of Eq. (7), this head is more precisely described as a weighted probability fusion of two branches of one network than as an ensemble of independently trained models; the term *Weighted Ensemble Head* is retained for consistency with the component naming used throughout.

J. Training Configuration

Training minimized a class-weighted cross-entropy loss with label smoothing, combined with a secondary cross-entropy term on the auxiliary branch. Writing $\tilde{y}_c = (1-\epsilon)\mathbb{1}[c = y] + \epsilon/C$ for the smoothed target with $\epsilon = 0.05$

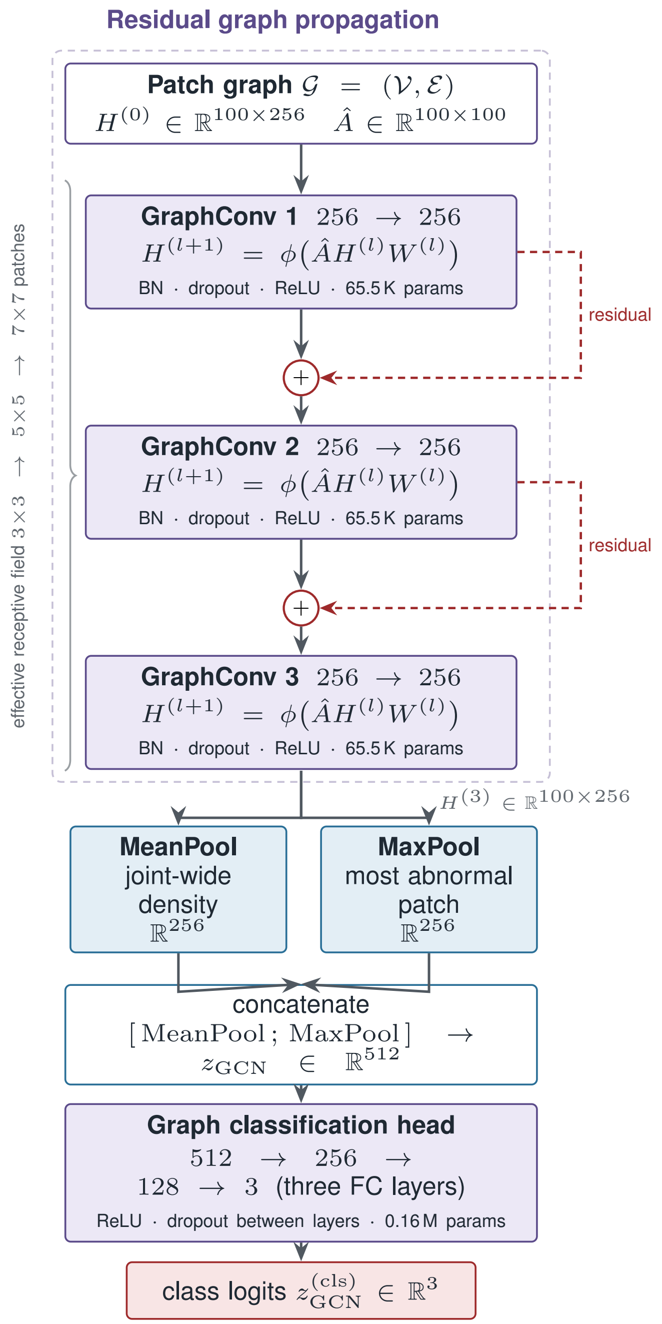

Fig. 5. Residual Dual-Pooling Graph Reasoning Branch. Three graph-convolutional layers (Eq. (4)), each $256 \to 256$ with batch normalization, dropout, and ReLU, are stacked with residual connections around layers 2 and 3. Stacking three eight-connected propagations widens the effective receptive field from 3×3 to 7×7 patches. The resulting node features $H^{(3)}$ are read out by concatenated mean–max pooling (Eq. (5)), giving a 512-dimensional embedding that passes through the three-layer classification head to the class logits $z^{(\mathrm{cls})}_{\mathrm{GCN}}$.

and $C = 3$, and ω_c for the inverse-frequency class weight of class c, the per-branch objective and total loss are

$$\mathcal{L}_{\mathrm{CE}}(z, y) = -\sum_{c=1}^{C} \omega_c\, \tilde{y}_c\, \log \mathrm{softmax}(z)_c,$$
$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(z^{(\mathrm{cls})}_{\mathrm{GCN}}, y) + \lambda_{\mathrm{aux}}\, \mathcal{L}_{\mathrm{CE}}(z_{\mathrm{CNN}}, y), \tag{7}$$

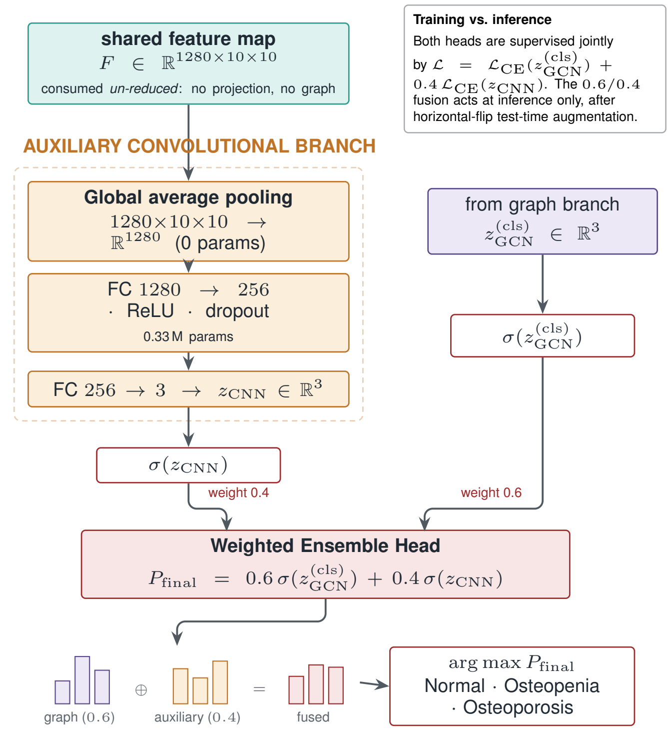

Fig. 6. Auxiliary Convolutional Branch and Weighted Ensemble Head. The auxiliary branch consumes the shared feature map F un-reduced, bypassing both the projection and the graph, and produces z_{CNN} through global average pooling and two fully connected layers. Its softmax output is combined with the graph branch's $z^{(\mathrm{cls})}_{\mathrm{GCN}}$ by the $0.6/0.4$ weighted probability fusion of Eq. (6), illustrated by the class-probability bars. Both heads are supervised jointly during training (Eq. (7)); the fusion itself acts only at inference, after horizontal-flip test-time augmentation.

with $\lambda_{\mathrm{aux}} = 0.4$. Because the training partition is already balanced by the augmentation of Section III-D, the inverse-frequency weights ω_c are uniform in practice; they are retained in the implementation so that the same objective can be applied unchanged to the unbalanced variant used in ablation (v).

Optimization used AdamW [46] with two parameter groups at differential learning rates: the unfrozen backbone layers were updated at 1×10^{-4}, while the projection layer, graph branch, and auxiliary branch were grouped together and updated more aggressively at 1×10^{-3}, both with a weight decay of 1×10^{-4}. A bare cosine schedule at the backbone learning rate destabilizes the pretrained weights in the first steps, so the learning-rate multiplier instead follows a linear warmup over the first three epochs followed by a single cosine decay to zero over the remaining 42 epochs, with no restarts:

$$\eta_t = \eta_{\mathrm{base}} \times \begin{cases} \dfrac{t+1}{T_{\mathrm{warm}}}, & t < T_{\mathrm{warm}} \\ \dfrac{1}{2}\left(1 + \cos\left(\pi\, \dfrac{t - T_{\mathrm{warm}}}{T_{\max} - T_{\mathrm{warm}}}\right)\right), & t \geq T_{\mathrm{warm}} \end{cases} \tag{8}$$

with $T_{\mathrm{warm}} = 3$ warmup epochs and $T_{\max} = 45$ total epochs. Gradients were clipped at a norm of 5.0 to guard against instability introduced by the graph convolutions. Training used mixed precision throughout except inside the graph branch, whose sparse scatter operations are run in full precision for numerical stability. Training was stopped early if validation

TABLE III
TRAINING AND INFERENCE CONFIGURATION

Input and backbone	
Input size	320×320
Backbone	EfficientNetB0 (last 6 of 9 blocks trainable)
Projection	1×1 conv, $1280 \rightarrow 256$
Graph structure	10×10 grid, 8-connectivity
Graph edges	684 directed (342 undirected)
GCN depth	3 layers, 256 hidden units
Optimization	
Optimizer	AdamW (differential learning rates)
Backbone LR / head LR	1×10^{-4} / 1×10^{-3}
Weight decay	1×10^{-4}
Scheduler	Warmup (3 ep.) + cosine decay, 45 ep.
Loss	Class-weighted CE, label smoothing 0.05
Auxiliary loss weight λ_{aux}	0.4
Batch size	24
Mixed precision	Enabled (except graph branch)
Gradient clip norm	5.0
Checkpoint selection	Best validation macro F1
Early-stopping patience	10 epochs
Random seed	**[TBD]**
Independent runs	**[TBD]**
Inference	
Fusion weight w	0.6 (GCN), 0.4 (CNN)
Test-time augmentation	Horizontal-flip averaging

macro F1-score, rather than validation accuracy, failed to improve for ten consecutive epochs; macro F1 is the model-selection criterion because the validation partition is naturally imbalanced (Section III-C), and accuracy alone can favour a model that neglects the minority Osteopenia class. At inference, predictions were averaged across the original image and its horizontal mirror, a lightweight test-time augmentation intended to reduce sensitivity to left–right orientation. Table III summarizes the configuration.

K. Experimental Setup

All experiments were implemented in Python 3.12 using PyTorch 2.11 and PyTorch Geometric (version **[TBD]**) for the graph-convolutional layers, OpenCV for image preprocessing, Albumentations [45] for augmentation, and scikit-learn for evaluation metrics. Training was carried out in a Google Colab GPU runtime on a single NVIDIA Tesla T4 with **[TBD]** GB of system RAM.

L. Evaluation Metrics and Statistical Analysis

Model performance was quantified on the test set using accuracy, macro-averaged precision, recall, and F1-score, together with a per-class classification report, a confusion matrix, and one-versus-rest ROC curves summarized by a macro-averaged AUC. All quantities follow their standard definitions as implemented in scikit-learn. Precision, recall, and F1 are reported as macro averages, weighting each class equally regardless of support, because the test partition is deliberately left at its natural, imbalanced distribution (Section III-C).

Because the severity grades are ordered rather than merely categorical, two ordinal measures are additionally reported. The quadratic weighted kappa compares the observed agreement matrix O against the agreement matrix E expected by chance, penalizing disagreements by the square of their grade distance:

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \qquad w_{ij} = \frac{(i-j)^2}{(C-1)^2}. \tag{9}$$

The mean absolute grade error, $\frac{1}{N}\sum_n |\hat{g}_n - g_n|$ over grades encoded as $0, 1, 2$, is reported alongside it. Both quantities distinguish a single-grade misclassification from the clinically far costlier Normal–Osteoporosis confusion, which nominal accuracy treats identically.

Interval estimates for accuracy and per-class recall are computed as Wilson score intervals at the 95% level, which is preferred to the normal approximation at the sample sizes involved here, particularly for the 57-image Osteopenia partition. Confidence intervals derived in this way from the reported counts are given in Section IV-B. Repeated-run variability, by contrast, cannot be estimated from a single training run: the results reported below come from one run with a single fixed partition, so no standard deviation across seeds or folds is available. The multi-seed and cross-validation protocol required to supply that estimate, together with paired significance testing against the baselines of Table VII, is reported as **[TBD]** pending completion and is stated as a limitation in Section VI.

M. Explainable Artificial Intelligence Techniques

Three explainability methods were applied to the trained model, each addressing a different level of granularity.

Grad-CAM [40] was computed over the final block of the EfficientNetB0 backbone using the graph branch's logits as the target signal. This construction differs from the standard application of Grad-CAM to a convolutional classifier: gradients propagate from $z_{\text{GCN}}^{(\text{cls})}$ back through the classification head, the three graph-convolutional layers, and the 1×1 projection before reaching the target activation map, so the resulting saliency reflects the graph branch's use of the backbone features rather than a purely convolutional decision path.

GNNExplainer [43] was applied to the trained graph head, learning a soft mask over node features to identify which of the one hundred patches most influenced a given prediction. Because the adjacency of Section III-G is fixed and identical for every input, the edge mask that GNNExplainer also produces is defined over a topology that does not vary across images and is therefore not informative for per-image comparison; only the node-feature mask is reported here.

LIME [41] was used to generate super-pixel-level explanations by perturbing the input image and observing the resulting change in the fused output probabilities of Eq. (6), isolating both the regions that supported a prediction and those that argued against it.

Feature-attribution methods such as SHAP [42] were considered as a fourth complementary view but were not included, since Shapley value estimation is substantially more expensive at the patch-graph resolution used here; extending the suite to include SHAP is left for future work.

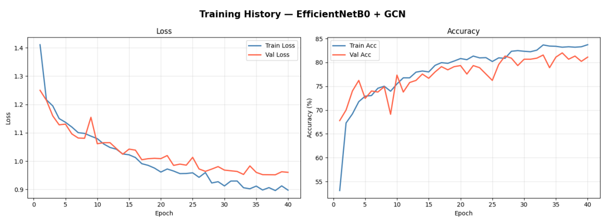

Fig. 7. Training and validation loss and accuracy curves.

IV. Results

This section reports training behaviour, test-set performance, discriminative power and error analysis, the ablation protocol, comparison with prior work, and explainability outcomes.

A. Training Behaviour

Training ran for a maximum of 45 epochs but was stopped early at epoch 41 after ten consecutive epochs without improvement in validation macro F1, the model-selection criterion (Section III-J). Training accuracy rose from 63.77% in epoch 1 to 93.30% at epoch 41, while validation accuracy rose from 69.18% to a peak of 87.33% at epoch 31, the checkpoint retained for final evaluation, at which validation macro F1 reached 85.96%. Training and validation loss declined together over the same span, from 1.21 to 0.45 on the training partition and from 0.98 to 0.66 on the validation partition, both measured at the retained epoch-31 checkpoint. Comparing the two accuracies at the final epoch gives a train–validation gap of 5.97 points, indicating mild but controlled overfitting rather than either severe underfitting or memorization. Fig. 7 depicts the training process.

B. Test-Set Performance

On the 293-image held-out test set, held at its natural, imbalanced class distribution (Section III-C), the EfficientNet-GCN model achieved an overall accuracy of 82.94%, with macro-averaged precision, recall, and F1-score of 0.8115, 0.8227, and 0.8161 respectively, as detailed in Table IV.

The 95% Wilson score interval for the overall accuracy is 78.21–86.81%, a span of roughly nine percentage points. This interval is wide because the test partition contains only 293 images, and it should be borne in mind whenever the headline figure is compared against values reported elsewhere. Per-class recall intervals, given in Table V, are wider still, and widest for Osteopenia, which is evaluated on 57 images.

The per-class breakdown (Table V) shows the strongest performance on the Normal class (precision 0.9107, recall 0.8718) and the weakest on Osteopenia (precision 0.7031, recall 0.7895), which is simultaneously the rarest class in the test set and, clinically, the intermediate grade whose radiographic appearance sits closest to both of its neighbours. Osteoporosis falls between the two (precision 0.8205, recall 0.8067). The pattern for Osteopenia, uniformly weaker precision and recall rather than a strength in one and a weakness in the other, is consistent with confusion concentrated at the two adjacent class boundaries rather than spread evenly across all class pairs.

TABLE IV
Overall Test Performance ($n = 293$)

Metric	Value
Accuracy	82.94% (95% CI 78.21–86.81)
Precision (macro)	0.8115
Recall (macro)	0.8227
F1-score (macro)	0.8161
AUC (macro, one-vs-rest)	0.9429
AUC (per class)	**[TBD]**
Top-2 accuracy	**[TBD]**
Quadratic weighted kappa	**[TBD]**
Mean absolute grade error	**[TBD]**
Expected calibration error	**[TBD]**

TABLE V
Per-class test performance, with 95% Wilson intervals on recall

Class	Prec.	Rec.	95% CI (rec.)	F1	Support
Normal	0.9107	0.8718	0.799–0.921	0.8908	117
Osteopenia	0.7031	0.7895	0.667–0.875	0.7438	57
Osteoporosis	0.8205	0.8067	0.727–0.868	0.8136	119
Macro avg	**0.8115**	**0.8227**	—	**0.8161**	293

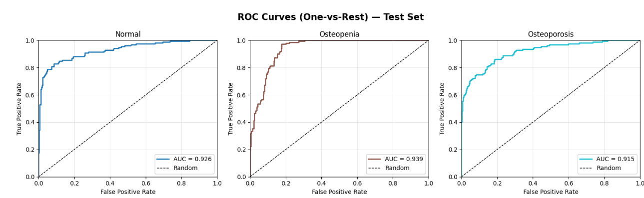

Fig. 8. One-vs-rest ROC curves per class, test set.

C. Discriminative Power and Error Analysis

One-vs-rest ROC analysis yielded a macro-averaged AUC of 0.9429 (Fig. 8), indicating strong class separability despite the more moderate hard-label accuracy. A gap of this size between ranking quality and thresholded accuracy is consistent with a model whose probability estimates order the three classes sensibly even where the arg-max decision is wrong. The top-2 accuracy that would quantify this directly is reported as **[TBD]**; until it is available the interpretation should be treated as a hypothesis rather than an established finding.

Fig. 9 details the test-set confusion matrix. Of the 293 test predictions, 243 were correct and 50 were errors. Of those 50, 31 fall on an adjacent-grade boundary (Normal–Osteopenia or Osteopenia–Osteoporosis), while 19 fall on the Normal–Osteoporosis boundary, the costliest possible error since it corresponds to a severe case read as entirely normal or the reverse. Adjacent-grade confusions are therefore the more common failure mode, but the 19 extreme-boundary errors represent 38% of all errors and 8% of the 236 Normal and Osteoporosis test images combined. This is not a negligible residue, and it is the single most important obstacle to clinical deployment of the model in its current form; it motivates the ordinal-modelling direction set out in Section VII.

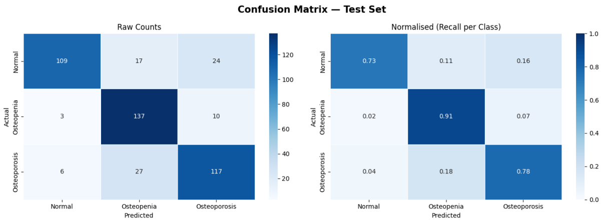

Fig. 9. Confusion matrix: raw counts and row-normalized recall, test set.

TABLE VI
ABLATION PROTOCOL AND RESULTS (TEST SET, $n = 293$)

Variant	Acc.	Macro F1	AUC
Full model (proposed)	82.94%	0.8161	0.9429
(i) w/o GCN branch	**[TBD]**	**[TBD]**	**[TBD]**
(ii) w/o auxiliary branch	**[TBD]**	**[TBD]**	**[TBD]**
(iii) w/o dual mean–max pooling	**[TBD]**	**[TBD]**	**[TBD]**
(iv) w/o CLAHE preprocessing	**[TBD]**	**[TBD]**	**[TBD]**
(v) w/o class-balancing augmentation	**[TBD]**	**[TBD]**	**[TBD]**
(vi) w/o test-time augmentation	**[TBD]**	**[TBD]**	**[TBD]**
(vii) GCN $\rightarrow$ param-matched conv	**[TBD]**	**[TBD]**	**[TBD]**

D. Ablation Study

To isolate the contribution of each architectural and methodological component, an ablation protocol was defined in which exactly one component of the pipeline is removed or substituted at a time, while the data splits (Table II), backbone, and training budget (Section III-J) are held fixed for every variant. Seven variants are defined: (i) *w/o GCN branch*, in which the patch graph and its three graph-convolutional layers are removed and the auxiliary branch alone produces the prediction; (ii) *w/o auxiliary branch*, in which the fusion of Eq. (6) is replaced by the graph branch's softmax output alone; (iii) *w/o dual pooling*, in which the concatenated mean–max readout is replaced by mean pooling only; (iv) *w/o CLAHE*, in which the contrast-enhancement step of Section III-B is skipped; (v) *w/o class-balancing augmentation*, in which the network is trained on the original 546/261/555 training distribution; (vi) *w/o test-time augmentation*, in which inference uses a single forward pass; and (vii) *GCN branch replaced by a parameter-matched convolutional block*, in which the graph branch is substituted by a stack of 3×3 convolutions with a comparable parameter budget operating on the same projected feature map F'.

Variant (vii) is included specifically to control for model capacity. Because the patch adjacency is fixed (Section III-G), the propagation of Eq. (4) aggregates over a 3×3 spatial neighbourhood at each layer, and any accuracy gain attributed to relational reasoning must be shown to exceed what an equally sized convolutional block achieves on the same features. Without this control, variant (i) alone cannot distinguish a benefit of graph structure from a benefit of added parameters.

Table VI lists these variants alongside the full model. At the time of writing only the full-model row reflects a completed training run, corresponding to the results reported in Table IV; the remaining runs are pending. The protocol is specified here in full so that each variant is reproducible from Section III, and the incompleteness is stated plainly rather than implied.

TABLE VII
BASELINES TRAINED UNDER THE IDENTICAL PROTOCOL OF SECTION III (TEST SET, $n = 293$)

Model	Acc.	Macro F1	AUC	Params
EfficientNetB0 + GAP + FC	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
ResNet50	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
DenseNet121	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
VGG19	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Sarhan et al. [19] (reimpl.)	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
EfficientNet-GCN (proposed)	**82.94%**	**0.8161**	**0.9429**	**5.03**

TABLE VIII
COMPUTATIONAL COST OF THE PROPOSED MODEL

Quantity	Value
Total parameters	5,027,970 (5.03M)
Trainable parameters	5,008,880 (99.6%)
Multiply–accumulate operations	**[TBD]**
GPU inference latency (T4, batch 1)	**[TBD]**
CPU inference latency (batch 1)	**[TBD]**
Peak inference memory	**[TBD]**

E. Baselines Under an Identical Protocol

Cross-study accuracy comparisons confound architecture with dataset, class count, and partitioning protocol. To separate these, Table VII reports established backbones trained and evaluated under exactly the preprocessing, partition, augmentation, and training budget described in Section III. These runs are pending at the time of writing.

F. Computational Efficiency

The proposed model contains 5,027,970 parameters (5.03M) in total, of which 5,008,880 (99.6%) are trainable under the partial fine-tuning scheme of Section III-F. This figure was obtained directly from the trained model rather than estimated, and breaks down as 4.01M in the EfficientNetB0 backbone, 0.33M in the projection layer, 0.36M in the graph branch, and 0.33M in the auxiliary branch. Because the framework is motivated by deployment in resource-constrained settings, parameter count alone is an insufficient measure of cost; Table VIII lists the throughput measurements needed to substantiate that motivation, which are pending.

G. Comparison with Prior Work

Table I situates the proposed framework against published deep-learning approaches to osteoporosis and BMD assessment across knee, hip, spine, dental, and CT imaging. Raw accuracy figures in that table should be read with caution rather than ranked directly: the compared studies differ in class count, imaging modality, dataset size, label source, and partitioning protocol, all of which affect the difficulty of the underlying decision problem. The proposed model's 82.94% accuracy carries a 95% confidence interval of 78.21–86.81% (Section IV-B), which overlaps several of the figures below it in the table.

Three comparisons within the three-class knee-radiograph setting warrant direct comment, since that is the setting most similar to the present one.

First, Sarhan et al. [19] report 97.50% on the same 1,947-image corpus, 14.6 points above the present result and the highest figure reported for this dataset. That study is also a three-class knee-radiograph formulation, so the difference cannot be attributed to an easier decision problem. One methodological difference is that the augmented corpus is partitioned after augmentation rather than before, which permits variants of a single radiograph to appear in both training and evaluation partitions; the present protocol partitions first (Section III-C). Whether this accounts for some, all, or none of the 14.6-point margin cannot be determined without reimplementing that pipeline under both orderings. That reimplementation is listed as a baseline in Table VII and is pending; until it is complete, no causal claim is made here, and the gap stands unexplained.

Second, Wani and Arora [5] report 91.10% on a three-class knee corpus, also above the present result, though on a substantially smaller cohort of 381 images, where the corresponding confidence interval would be wider still.

Third, the present result is comparable to Naguib et al.'s [25] 83.74% on a 611-image knee corpus, and exceeds Feng et al.'s [18] 74.00% on a 139-image hip cohort.

The proposed framework therefore does not lead this literature on accuracy, and is not claimed to. Its contribution is instead threefold and orthogonal to the accuracy ranking: an explicit relational representation over patch-level features rather than a purely Euclidean one; a partition-before-augment protocol that removes a specific and widely present source of optimistic bias; and the joint reporting of three complementary explanation views alongside a full parameter budget, in a literature where no reviewed study reports a parameter count at all. Whether any reviewed study reports more than a single explanation method is recorded in the final column of Table I. Establishing that the relational component is responsible for any part of the performance obtained requires the ablation of Table VI and the baselines of Table VII, both pending.

H. Explainability Results

Grad-CAM, computed on the final EfficientNetB0 block, localized activations over cortical and medullary bone regions for the correctly classified Osteopenia and Osteoporosis cases examined, rather than diffusing over background or soft tissue (Fig. 10). GNNExplainer, applied to the graph head, identified a small, spatially contiguous cluster of patch-graph nodes as the dominant contributors to each prediction examined (Fig. 11). LIME's super-pixel explanations (Fig. 12) were broadly consistent with the Grad-CAM findings, with positive regions concentrating over trabecular and cortical structures for both Osteopenia and Osteoporosis predictions.

Two qualifications apply to these observations. First, they are qualitative assessments of a selected set of correctly classified cases; no quantitative faithfulness measure was computed, and no explanations of misclassified cases were examined, so the evidence available cannot exclude the possibility that the same methods would produce equally plausible-looking maps for incorrect predictions. Deletion and insertion AUC scores for each of the three methods, together with explanations of

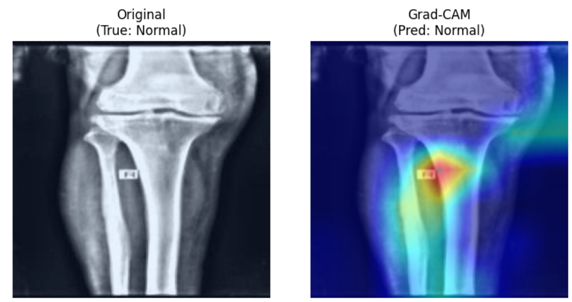

Fig. 10. Grad-CAM activation overlays on representative test samples.

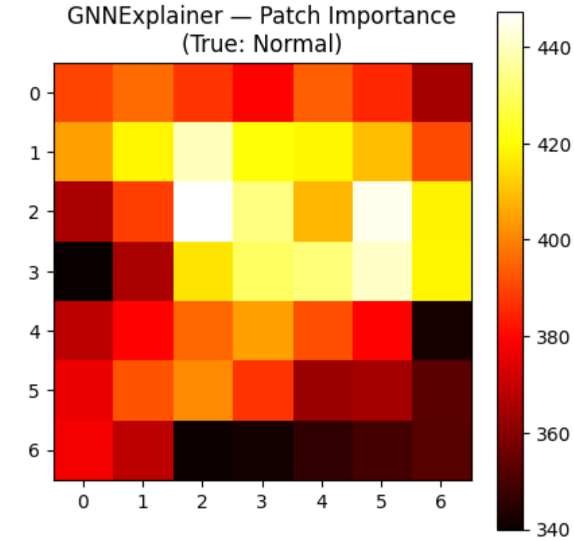

Fig. 11. GNNExplainer patch-importance heatmap and overlay on the original radiograph.

representative failure cases, are reported as **[TBD]**. Second, no clinician has reviewed the explanations, so their agreement with expert judgement of which regions carry diagnostic signal is untested. Accordingly, the results in this section are stated as consistency with anatomically plausible regions, not as confirmation that the model reasons clinically.

V. Discussion

A. Where the Model Fails, and Why

The error structure reported in Section IV is coherent with the clinical structure of the task. Osteopenia is both the rarest class in the test partition and the intermediate grade, defined by a T-score band lying between the other two, and it is the class on which the model is weakest in both precision and recall. A classifier trained with a nominal cross-entropy objective treats the three grades as unordered categories, so it receives no signal that confusing Normal with Osteoporosis is worse than confusing either with Osteopenia. The 19 extreme-boundary errors observed are the direct consequence of that modelling choice. This suggests the task is better posed as an ordinal regression problem than as a three-way nominal classification, a direction taken up in Section VII.

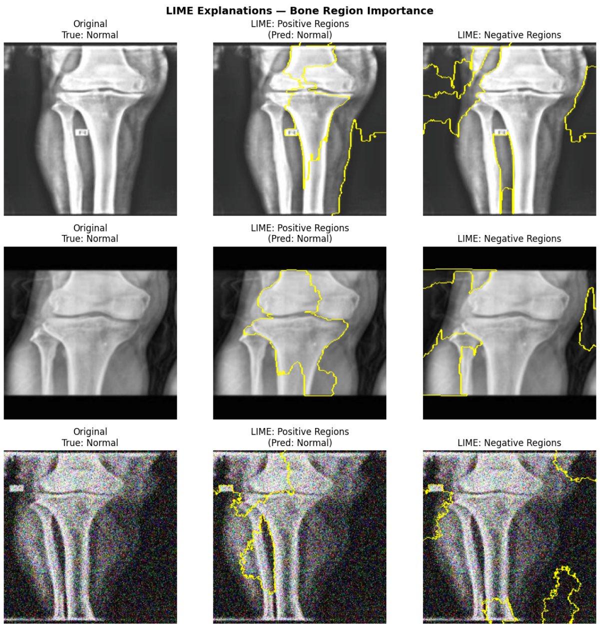

Fig. 12. LIME positive and negative super-pixel explanations across three test samples.

B. Does the Patch Graph Do Anything a Convolution Cannot?

The most substantial theoretical objection to the present design concerns the fixed adjacency. Because A is an eight-connected lattice identical for every input image, the propagation of Eq. (4) computes, at each node, a degree-normalized sum over that node's 3×3 spatial neighbourhood followed by a shared linear map. This is structurally close to a 3×3 convolution whose kernel weights are fixed and tied across positions rather than learned, and the objection follows that the graph branch may be reproducing an operation the convolutional backbone already performs, with the apparent benefit coming from the additional 0.36M parameters rather than from relational reasoning.

Three properties distinguish the two operations. The symmetric normalization $\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$ makes each node's aggregation degree-dependent, so patches at the grid boundary and corners, which have five and three neighbours respectively rather than eight, are weighted differently from interior patches, whereas a convolution applies an identical kernel everywhere and handles boundaries by padding. Stacking three such layers yields an effective receptive field spanning a 7×7 patch region, which at this feature-map scale covers a substantial fraction of the joint. And the formulation generalizes directly to a learned or feature-dependent adjacency, which a fixed convolution does not.

These are arguments for a difference in principle. They are not evidence of a difference in effect on this task, and the present results do not supply that evidence: variant (vii) of Table VI is precisely the experiment that would settle it, and it is pending. Until it is run, the claim that relational reasoning contributes to the reported performance should be regarded as motivated but unverified. This is stated explicitly because it is the manuscript's principal methodological vulnerability, and because the alternative, a learned adjacency of the kind used by MedGNN [39] or an attention-weighted formulation [37], would make the relational component parametrically irreducible to a convolution and is the natural next step (Section VII).

C. Ranking Quality Versus Decision Quality

The macro AUC of 0.9429 alongside 82.94% accuracy indicates that the model's class rankings are considerably better than its arg-max decisions. Two non-exclusive explanations are available: the decision rule may be miscalibrated, in which case threshold adjustment on the validation partition would recover accuracy without retraining, or the fixed fusion weight $w = 0.6$ of Eq. (6) may be suboptimal, since it was set a priori rather than tuned. Both are cheap to test and neither has been tested here. The calibration error and the fusion-weight sweep reported as **[TBD]** would distinguish them.

D. Clinical Positioning

The framework is positioned as a triage aid rather than a diagnostic replacement. In a screening pathway, its plausible role is to flag radiographs warranting DXA referral, a binary decision, rather than to assign a definitive severity grade. Reported three-class accuracy is not the metric that governs performance in that role; sensitivity at a clinically acceptable specificity is. The present evaluation does not report an operating-point analysis of this kind, and adding one is a priority (Section VII). Any deployment claim would additionally require the external validation the study lacks.

VI. Limitations

Seven limitations qualify the results reported above.

First, and most importantly for interpreting Table VI, seven of the eight ablation variants remain unrun. The protocol is fully specified in Section IV-D, but only the full model's row reflects a completed training run, so no component of the architecture has yet been shown to contribute to the reported performance.

Second, no baseline has been trained under the identical protocol (Table VII). The comparison in Section IV-G is therefore against figures obtained on different partitions and, in most cases, different datasets, which does not isolate the effect of the proposed architecture.

Third, the reported metrics come from a single training run on a single fixed partition. No repeated-seed or cross-validation estimate of variance is available, and no significance test against any alternative has been performed. The Wilson intervals in Section IV-B capture sampling uncertainty in the test partition but not variability across training runs or partitions.

Fourth, consistent with Gap 5 of Section II-E, the framework has been evaluated on a single publicly distributed dataset [9] and has not been tested on independent, multi-centre clinical data, so its generalization across scanners and acquisition protocols is unverified.

Fifth, the source dataset provides no patient-level identifiers (Section III-A). The partition was verified free of image-level overlap, but if the corpus contains bilateral pairs or repeat studies from the same individual, patient-level leakage cannot be excluded and the reported metrics would be optimistic. This limitation applies to any study using this corpus, including those the present work compares against.

Sixth, because the validation and test partitions are deliberately left at their natural distribution, the Osteopenia class is evaluated on only 57 images, and its recall interval spans 0.667–0.875 (Table V). Conclusions about this class are correspondingly weak.

Seventh, radiographic severity labels were taken as provided by the source dataset rather than confirmed against DXA-derived T-scores, so label noise at the class boundaries the model confuses most often cannot be ruled out. Several design choices, including the 320×320 input resolution, the random-erasing augmentation, and the fusion weight $w = 0.6$, were also fixed a priori rather than selected empirically.

VII. Future Work

Completing the ablation of Table VI, including the parameter-matched convolutional control, and the identically trained baselines of Table VII is the most immediate priority, since together they determine whether the architectural claims of this paper hold. Repeating the protocol across multiple seeds or cross-validation folds, with paired significance testing, would supply the variance estimates the present results lack.

Beyond this, five directions follow. Reformulating the task as ordinal regression, through a CORAL-style cumulative-link head or a distance-weighted loss, directly targets the extreme-boundary errors identified in Section V and matches the ordered structure of the severity scale. Replacing the fixed eight-connectivity adjacency with a learned edge-weighting scheme, following the attention-based formulation [37] or the k-nearest-neighbour construction of MedGNN [39], would make the relational component parametrically distinct from a convolution. Adding quantitative explanation-faithfulness measures and radiologist review would convert the qualitative explainability findings into evidence. Reporting a binary DXA-referral operating point, with sensitivity at a fixed specificity, would evaluate the model in the clinical role actually proposed for it. Finally, external multi-centre validation and the collection of additional Osteopenia cases would address the two gaps this study inherits from the wider literature.

VIII. Conclusion

This paper presented EfficientNet-GCN, a hybrid convolutional–graph framework for osteoporosis severity classification from knee radiographs, built around a Patch-Graph Construction Module, a Residual Dual-Pooling Graph Reasoning Branch, and an Auxiliary Convolutional Branch fused through a Weighted Ensemble Head. An ImageNet-pretrained EfficientNetB0 encoder extracts spatial features; the Patch-Graph Construction Module channel-reduces and reformulates them into a 100-node, eight-connected graph; and the graph branch propagates relational context across that graph, treating anatomically adjacent regions of the knee joint as connected nodes rather than isolated units.

The corpus is partitioned before any augmentation is applied and only the training partition is rebalanced, so the validation and test partitions retain their natural class distribution and no augmented variant of a test radiograph can appear in training. Under this protocol the model reaches 82.94% accuracy (95% CI 78.21–86.81%) and a macro one-vs-rest AUC of 0.9429 on the held-out test set, with 5.03M parameters. Grad-CAM, GNNExplainer, and LIME localize predictions to cortical and trabecular bone regions in the cases examined.

The framework does not exceed the best previously reported accuracy on this corpus, and the present evidence does not yet establish that its relational component is responsible for the performance obtained: the ablation of Table VI, the parameter-matched convolutional control, and the identically trained baselines of Table VII are all pending, and Section V sets out why the last of these is decisive. What the study does contribute is a leakage-controlled evaluation protocol, an explicit relational representation over patch-level features, and the joint reporting of three complementary explanation views alongside a full parameter budget, in a literature where at most one explanation method and no parameter count are typically reported. Sections VI and VII state the remaining gaps and the steps planned to close them.

Data and Code Availability

The knee radiograph dataset used in this study, the Multi-Class Knee Osteoporosis X-Ray Dataset, is publicly available on Kaggle [9] at https://www.kaggle.com/datasets/mohamedgobara/multi-class-knee-osteoporosis-x-ray-dataset. Source code, the partition manifest, and the random seeds required to reproduce the reported results are available at **[TBD]**.

Ethical Statement

This study used a publicly available, de-identified, third-party imaging dataset [9] and involved no prospective recruitment, no intervention, and no access to identifiable patient information. Institutional review board approval and informed consent were therefore not required. The authors did not attempt to re-identify any individual represented in the data.

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

T. S. Sintheia: conceptualization, methodology, software, investigation, writing—original draft. M. R. Muin: software, validation, data curation, visualization. P. D. Gupta: investigation, formal analysis, visualization. M. I. Mobin: supervision, methodology, writing—review and editing. All authors read and approved the final manuscript.

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